Prospective Study of Homologous Recombination Repair Gene Mutation Prevalence in Patients With Advanced Prostate Cancer From Latin America: Challenges and Future Approaches

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ABSTRACT

PURPOSE The prevalence of homologous recombination repair gene mutations (HRRm) in patients with metastatic castration-resistant prostate cancer (mCRPC) in Latin America and the Caribbean (LAC) is unknown. Prevalence of homologous Recombination repair (HRR) gene mutatiOns in patientS with metastatic castration resistant ProstatE Cancer in LaTin America (PROSPECT) aimed to determine this prevalence and to describe the demographic and clinical characteristics of the participants.

MATERIALS AND METHODS This was a prospective, cross-sectional, multicenter study across 11 cancer centers in seven LAC countries. After informed consent, all eligible participants underwent genomic testing by provided blood samples for germline HRR testing; they also provided PC tissue blocks if available for somatic HRR testing.

RESULTS Between April 2021 and April 2022, 387 patients (median age, 70 years [49-89], 94.3% Eastern Cooperative Oncology Group 0-1) with mCRPC were enrolled in the study. Almost 40% of them had a family history of cancer, and the overall time from their initial PC and mCRPC diagnosis was 3 years and 1 year, respectively. The overall prevalence of germline HRRm was 4.2%. The mutations detected included the genes *CHEK2* (n = 4, 1%), *ATM* (n = 3, 0.8%), *BRCA2* (n = 3, 0.8%), *BRIP1* (n = 2, 0.5%), RAD51B (n = 2, 0.5%), BRCA1 (n = 1, 0.3%), and MRE11 (n = 1, 0.3%). The prevalence of somatic HRRm could not be assessed because of high HRR testing failure rates (79%, 199/251) associated with insufficient DNA, absence of tumor cells, and poor-quality DNA.

CONCLUSION Despite the study's limitations, to our knowledge, PROSPECT was the first attempt to describe the prevalence of HRRm in patients with PC from LAC. Notably, the germline HRRm prevalence in this study was inferior to that observed in North American and European populations. The somatic HRR testing barriers identified are being addressed by several projects to improve access to HRR testing and biomarker-based therapies in LAC.

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INTRODUCTION

Prostate cancer (PC) is the most common oncologic diagnosis in Latin America and the Caribbean (LAC) among men with 214,522 new cases in 2020.[1] In this region, it represents 15.2% of all cancers and it constitutes almost a third (29.8%) of all the incident cancer cases in men.[1] Although the reported incidence of PC in LAC is slightly lower than in North America (NA; 16.9%), its mortality rate is approximately 50% higher (LAC, 15.3%; NA, 9.9%).[1] Moreover, LAC's population will age more rapidly than North America's population, with an expected increase from 2019 to 2050 of 156% compared with 48% in NA.[2] The combination of higher PC mortality rates along with the increased risk of PC with advancing age[3] and LAC's rapidly aging population plus metastatic PC's poor survival rate (32% at 5 years)[4] and its higher health care costs require a better characterization of advanced PC in LAC. Unequal access to life-prolonging therapies and treatment-monitoring technologies, inadequate health care budgets, fragmented health care systems, and resources concentrated in urban areas in LAC countries[5] highlight the need to better describe this population since the published literature on

CONTEXT

Key Objective
The prevalence of homologous recombination repair (HRR) gene mutations in patients with metastatic castration-resistant prostate cancer (mCRPC) in Latin America and the Caribbean (LAC) is unknown. Prevalence of homologous Recombination repair (HRR) gene mutatiOns in patientS with metastatic castration resistant ProstatE Cancer in LaTin America aimed to determine this prevalence and to describe the demographic and clinical characteristics of patients with advanced PC from LAC, a largely underserved population.

Knowledge Generated
In this prospective study, germline HRR alterations were found at a lower rate than what has been observed in American and European cohorts. Viable tumor archival tissue was the main systematic barrier to tissue-based somatic testing in LAC.

Relevance
The lower rates of germline HRR alterations impact the systemic treatment options available to mCRPC patients and the process of cascade genetic testing of their family members. As a result of this study, several ongoing initiatives to improve access to HRR testing for patients with mCRPC in LAC were implemented.

this topic is lacking, especially regarding the metastatic stage.

Homologous recombination repair (HRR) is an accurate multistep process to repair DNA double-stranded breaks that is necessary to preserve genomic stability.[6,7] Mutations in the HRR (HRRm) pathway are associated with many cancers, particularly those of the breast, ovary, and prostate.[7] These mutations have a prognostic and predictive value, and they are relatively common (11.8%-28.4%) in patients with metastatic castration-resistant PC (mCRPC).[8-10] For patients with mCRPC and HRR genomic defects, targeted therapies have been shown to improve their clinical outcomes.[10-15] But despite these improvements, the genomic sequencing of PC in LAC lags behind the progress made in breast, lung, colon, gastric, and ovarian cancers in the region.[16,17]

Recognizing the importance of better understand the genomic testing rate patterns among LAC countries, we launched Prevalence of homologous Recombination repair (HRR) gene mutatiOns in patientS with metastatic castration resistant ProstatE Cancer in LaTin America (PROSPECT) to prospectively determine the prevalence of HRRm in patients with mCRPC in LAC. Additional exploratory objectives were to describe the demographic and clinical characteristics and associations between these characteristics and the prevalence of HRRm in the study participants.

MATERIALS AND METHODS

Study Design

PROSPECT was a prospective, cross-sectional, noninterventional, multicenter study to determine the prevalence of HRRm in participants with mCRPC in Latin American countries.

This study took place in 11 reference centers for the treatment of patients with mCRPC in LATAM countries (Argentina [two], Brazil [three], Colombia [one], Costa Rica [one], Mexico [two], Panama [one], and Peru [one]). The maximum duration of the study was expected to be 12 months with planned study dates from February 2021 to January 2022. The actual study period was from April 2021 to April 2022 and planned to enroll a minimum of 340 patients.

Eligible participants had confirmed mCRPC, age 18 years and older, and were able to provide medical and demographic information, and blood samples. If available, tissue samples were to be submitted as well. Candidates were not included in the study if they had a diagnosis of any severe acute or chronic medical or psychiatric condition that could increase the risk associated with study participation or may interfere with test result interpretation or if they were unable to provide their informed consent to participate.

Both the blood samples and tissue blocks were analyzed at local laboratories in Argentina and Brazil and at a central laboratory in Mexico for those originating from Colombia, Costa Rica, Mexico, Panama, and Peru. The research-use-only AmoyDx Halo-shape Annealing and Defer-Ligation Enrichment (HANDLE) HRR next-generation sequencing panel was used to analyze tissue and blood for somatic and germline mutations, respectively. The germline assays included HRR and other genes of interest. This panel is able to detect single-nucleotide variants (SNVs), and insertion and deletion variants (Indels) in protein-coding regions and intron/exon boundaries in 25 HRR genes (*AR*, *ATM*, *ATR*, *BARD1*, *BRCA1*, *BRCA2*, *BRIP1*, *CDH1*, *CDK12*, *CHEK1*, *CHEK2*, *FANCA*, *FANCL*, *HDAC2*, *MRE11*, *NBN*, *PALB2*, *PPP2R2A*, *PTEN*, *RAD51B*, *RAD51C*, *RAD51D*, *RAD54L*, *STK11*, and *TP53*) plus *HOXB13* and *ESR1*. It also detects SNVs and Indels in the hotspot regions of five driver genes (*BRAF*, *ERBB2*, *KRAS*, *NRAS*, and *PIK3CA*) and it detects large rearrangements in *BRCA1/2* genes from blood-derived DNA. Test result positivity was defined by any genomic alterations in one of the genomic alterations tested by AmoyDx HANDLE.

All the samples were labeled and shipped in accordance with applicable local laws and regulations, and the complete instructions for sample processing, handling, and shipment were followed according to the manufacturer. In addition, a full chain of custody was maintained for them throughout the study.

The investigators at each site were responsible for notifying participants of their HRR mutation status, either directly or through their treating physician, according to the local standard of care.

Data Collection

This study was conducted in compliance with each country's Ministry of Health regulatory requirements and the ethical principles from the Helsinki Declaration. The protocol and any amendments to it were submitted for ethical review, and approval was obtained in writing from each investigator's institutional review board.

Data were handled in accordance with Good Clinical Practice, federal, and local regulations. All source documents were completed by the site personnel or study staff and signed by the data collector. The source documents were reviewed, signed, and dated by the investigators.

During screening, patients with confirmed mCRPC fulfilling the eligibility criteria were identified and invited to participate in the study. Demographic and clinical data were collected in case report forms and blood samples were collected from all eligible participants during the enrollment process. Archived formalin-fixed paraffin-embedded (FFPE) tissue blocks were requested from the corresponding pathology laboratory for all eligible participants. Eligible FFPE blocks were ≤5 years old.

Statistical Design

The calculated sample size was 340 patients. It was anticipated that patients would be enrolled from multiple sites, including Argentina, Brazil, Colombia, Costa Rica, Mexico, Panama and Peru. Panama, Costa Rica, Colombia, and Peru were considered a cluster. The sample size was estimated with a confidence level of 95%, with a margin of error of 5% and with a projected HRR mutation mCRPC prevalence of 28% according to a previous report.[16] The final data (February 23, 2023) were analyzed by descriptive statistics, mainly by absolute and relative frequencies. Demographic information as well as clinical, general pathologic, and hereditary background characteristics were summarized using descriptive statistics. HRR gene somatic and germline mutation prevalence were, respectively, determined in tissue and in blood on the basis of all eligible cases. Specific gene alterations were determined on the positive cases with HRR alterations. All estimations were performed with 95% CIs as implemented in the SAS software (version 9.4, SAS Institute Inc, Cary, NC). Associations between baseline characteristics (age, country, family cancer history, medical history of cancer, location of metastases, and Eastern Cooperative Oncology Group [ECOG] status) with HRR germline mutations were analyzed by using Cramer's V to measure the strength of association between two nominal variables.

Statement of Ethics

The authors state that every effort was made to follow all the local and international ethical guidelines and laws that applied to this research.

RESULTS

From April 2021 to April 2022, 395 patients were screened and 387 patients were enrolled in the study from Argentina (n = 67), Brazil (n = 36), Colombia (n = 83), Costa Rica (n = 35), Mexico (n = 90), Panama (n = 59), and Peru (n = 17) (Fig 1). Dropouts were not observed in this study. Missing data were not imputed.

Table 1 summarizes baseline characteristics of the study population. The median age was 70 years and most (94%) of the patients had an ECOG performance status (ECOG PS) of 0 and 1. Overall, over a third (151, 39%) of the patients had a history of cancer in their family, the majority in first-degree relatives (120, 80%) and some in second-degree relatives (31, 20%); the most commonly reported cancers among family members were prostate (53%) and breast (21%); the countries with the highest number of patients with family cancer history were Costa Rica (68.6%), Brazil (66.7%), and Panama (66.1%). Notably, for 11.9% of the patients, these data were unknown.

More than two thirds of patients (70%) had no previous local therapy, and the median time elapsed since the initial PC diagnosis until genetic testing was 3 years (0-23 years) and the time from mCRPC to genetic testing was 1 year (0-7), without major differences among the six countries that participated in the study (Table 1).

Overall, the most common location of the metastases was bone only (220/387, 57%), followed by lymph nodes (54/387, 14%) and visceral (9/387, 2%; lung, five; liver, two; other, two). Almost a quarter of the participants (86/387, 22%) had metastases in multiple sites. This distribution was relatively consistent in Argentina, Colombia, Costa Rica, Mexico, and Panama. In Brazil and Peru, the distribution was different: the most common location of metastases was lymph nodes. However, these data were missing for the other half (n = 18) of the patients from Brazil.

Most of the blood samples for HRR germline testing arrived at the laboratories and were analyzed (98%, n = 379). Among the 2% (n = 8) that were not analyzed, the reasons are shown in Figure 1. The overall prevalence of HRR germline mutations was 4.27% (n = 20; median, 4; range, 2-6; Table 2). The countries with the highest rates were Mexico (6.6%), Costa Rica (6.3%), and Argentina (6.2%). The five most frequent HRR germline mutations were in *CHEK2* (n = 4), *ATM* (n = 3), *BRCA2* (n = 3), *BRIP1* (n = 2), and *RAD51B* (n = 2; Table 2). There was one positive result for *BRCA1* and *MRE11*. A third (six of 20, 30%) of these participants reported having first-degree relatives with history of cancer, the rest did not report any family history of cancer.

There were no statistically significant associations between baseline characteristics of interest and HRR germline alterations (Table 3). Weak positive associations with age, ECOG PS, and country were found.

Analyses to detect germline variants of uncertain significance (VUS) were also performed in this study. The most frequent ones were *BRCA2* (5.4%, n = 21), *FANCA* (4.7%, n = 18), *ATM* (4.1%, n = 16), *ATR* (3.9%, n = 15), *CHEK2* (3.1%, n = 12), *MRE11* (3.1%, n = 12), *PALB2* (2.8%, n = 11), *BRCA1* (2.3%, n = 9), *BRIP1* (2.3%, n = 9), *FANCL* (2.3%, n = 9), *RAD51B* (2.1%, n = 8), and *CDK12* (2.1%, n = 8).

Almost two thirds (65%, n = 251) of the patients had FFPE tissue blocks available (Fig 1). The countries that submitted most samples were Colombia (n = 83), Mexico (n = 47), Panama (n = 44), and Argentina (n = 41); Brazil (n = 19), Costa Rica (n = 10), and Peru (n = 6) contributed few

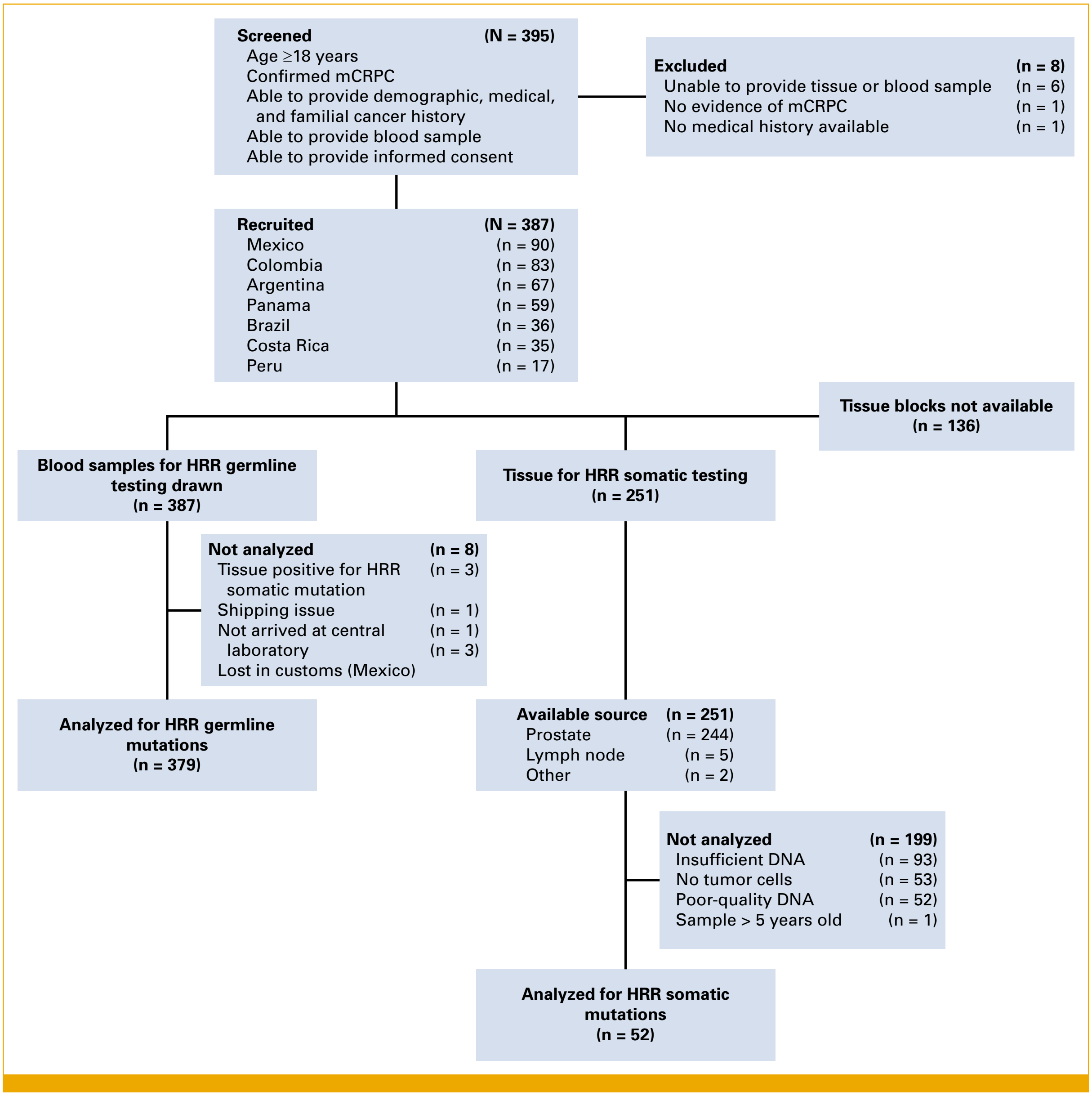

FIG 1. CONSORT diagram. HRR, homologous recombination repair; mCRPC, metastatic castration-resistant prostate cancer.

TABLE 1. Baseline Patient and Tumor Characteristics

Characteristics	Argentina	Brazil	Colombia	Costa Rica	Mexico	Panama	Peru	Overall
Total patients, No. (%)	67 (17.3)	36 (9.3)	83 (21.4)	35 (9)	90 (23.3)	59 (15.2)	17 (4.4)	387
Age, years, median (range)	72 (57-87)	68 (54-85)	72 (49-89)	69 (51-89)	68 (53-89)	71 (53-89)	70 (62-89)	70 (49-89)
ECOG PS, No. (%)								
0-1	67 (100)	36 (100)	83 (100)	30 (85.7)	74 (82.2)	59 (100)	16 (94.1)	365 (94.3)
2				5 (14.3)	11 (12.2)		1 (5.9)	17 (4.4)
Family history of cancer, No. (%)								
Yes	9 (13.4)	24 (66.7)	25 (30.1)	24 (68.6)	24 (26.7)	39 (66.1)	6 (35.3)	151 (39)
No	24 (35.8)	10 (27.8)	56 (67.5)	10 (28.6)	60 (66.7)	19 (32.2)	11 (64.7)	190 (49.1)
Unknown	34 (50.7)	2 (5.6)	2 (2.4)	1 (2.9)	6 (6.7)	1 (1.7)	0	46 (11.9)
Years since PC diagnosis, median (range)	4 (0-23)	4.5 (1-22)	2 (0-5)	4 (1-19)	2 (0-19)	2 (0-13)	4 (0-9)	3 (0-23)
Years since mCRPC diagnosis, median (range)	1 (0-5)	1 (0-5)	1 (0-4)	1 (0-7)	0 (0-6)	0 (0-6)	2 (0-7)	1 (0-7)
Previous local therapy, No. (%)								
Surgery	9 (13)	3 (8)		6 (17)	5 (5)	2 (3)	1 (6)	27 (7)
Radiation therapy	21 (31)	2 (5)	4 (5)	6 (17)	11 (12)	1 (2)	2 (12)	47 (12)
Missing	0	21 (58)	0	0	0	0	0	21 (5)
Metastases location, No. (%)								
Bone	40 (60)	0 (0)	51 (61)	21 (60)	60 (67)	42 (71)	6 (35)	220 (57)
Lymph nodes	10 (15)	16 (44)	13 (16)	4 (11)	3 (3)	1 (2)	7 (41)	54 (14)
Multiple sites[a]	16 (24)	2 (6)	16 (19)	9 (26)	25 (28)	14 (24)	4 (24)	86 (22)
Visceral	1 (1)	0	3 (4)	1 (3)	2 (2)	2 (3)	0	9 (2)
Missing	0	18 (50)	0	0	0	0	0	18 (4.7)

Abbreviations: ECOG PS, Eastern Cooperative Oncology Group performance status; mCRPC, metastatic castration-resistant prostate cancer; PC, prostate cancer.
[a]Bone plus lymph nodes or visceral metastases.

samples. Although most blocks were stored for 4 years or less (84.5%), their shelf-life varied considerably depending on the country. The most common tissue of origin for the FFPE blocks was the prostate (97%, n = 244), but for some patients, they originated from lymph nodes or other locations.

The rate of unavailable FFPE samples was high, with a median of 44% (0%-71%). Colombia was the only country from which all patients had tissue samples available; the percentage of unavailable tissue blocks was the highest in Costa Rica (71%), Peru (65%), and Mexico (48%).

As shown in Figure 1, analytic results from tissue blocks were reported for 251 patients. However, only 20.7% of them (n = 52/251) were appropriate for HRR testing. The median number of viable blocks obtained per country was 5 (range, 1-20). The countries with the highest rates of viable tissue blocks were Panama (46%, n = 20) and Mexico (29%, n = 15). Out of the overall cohort (N = 387), over a third of the participants had no tissue blocks available (35.1%, n = 136/387) and over half of them (51.4%, n = 199/387) could not be analyzed for somatic mutations. The most common reasons for these HRR testing failures were insufficient DNA (47%, 93/199), no tumor cells (27%, 53/199), and poor-quality DNA (21%, 52/199). The reasons varied by country. For instance, in Argentina, the most common problem was insufficient DNA, while in Colombia, it was no tumor cells, and in Brazil and Mexico, it was poor-quality DNA. Since most of the data (86.6%, 335/387) from tissue FFPEs to test for somatic HRR mutations were missing either because tissue was not available or because of HRR testing failure, we are not reporting the results for the small number of samples that were analyzed (13.4%, n = 52/387).

DISCUSSION

In this study, we prospectively assessed the genetic testing rates in a large cohort of patients from seven LAC. Considering 39% of the participants had family history of cancer, the prevalence of HRR germline mutations was low at 4.2%. The rate of tissue-based somatic testing rates was unexpectedly low, mostly because of limitations associated with tissue collection and low-quality or insufficient DNA.

These mutations have a prognostic and predictive value and they are relatively common (11.8%-28.4%) in patients with mCRPC.[8-10] For patients with mCRPC and HRR genomic defects, targeted therapies have been shown to improve their clinical outcomes.[10-15]

The identification of HRR gene defects has emerged as an important component in the management of advanced PC,

TABLE 2. Prevalence of HRR and Other Germline Mutations in Blood Samples by Country, by Mutation, and Overall

Country	ATM, No. (%)	*BRCA1*, No. (%)	*BRCA2*, No. (%)	*BRIP1*, No. (%)	*CHEK2*, No. (%)	*MRE11*, No. (%)	*RAD51B*, No. (%)	HRR Total, No. (%)	*ESR1*, No. (%)	*HOXB13*, No. (%)	Germline Mutation Total, No. (%)
Argentina[a] (n = 65)											
Positive	0	0	2 (3)	1 (1.5)	1 (1.5)	0	0	4 (6.2)	0	1 (1.5)	5 (7.7)
Variants	—	—	NM_000059.3 c.6405_6409del, deletion exon 1 (NM_000059.3 ex1del)	NM_032043.3 c.1066C>T	NM_007194.3 c.793-1G>A	—	—	—	—	NM_006361.5 c.251G>A	
Brazil[a] (n = 35)											
Positive	0	0	0	0	0	1 (2.8)	0	1 (2.8)	1 (2.8)	2 (5.6)	4 (11.4)
Variants	—	—	—	—	—	NM_005591.3 c.1516G>T	—	—	NM_001122740.1 c.896A>G	NM_006361.5 c.251G>A, NM_006361.5 c.251G>A	
Colombia[a] (n = 81)											
Positive	0	0	0	0	0	0	0	0	0	0	0
Variants	—	—	—	—	—	—	—	—	—	—	
Costa Rica[a] (n = 32)											
Positive	1 (2.9)	0	0	0	0	0	1 (2.8)	2 (6.3)	0	0	2 (6.3)
Variants	NM_000051.3: c.5908C>T	—	—	—	—	—	NM_133509.4:c.198+1G>T	—	—	—	
Mexico (n = 90)											
Positive	1 (1.1)	0	1 (1.1)	1 (1.1)	2 (2.2)	0	1 (1.1)	6 (6.6)	0	0	6 (6.7)
Variants	NM_000051.3: c.382del	—	NM_000059.3:c.3264dup	NM_032043.3:c.3730_3731del	NM_001005735.1: c.836T>C, NM_001005735.1: c.836T>C	—	NM_133509.4:c.452+3A>G	—	—	—	
Panama (n = 59)											
Positive	1 (1.7)	1 (1.7)	0	0	1 (1.7)	0	0	3 (5.1)	0	0	3 (5.1)
Variants	NM_000051.3: c.1795dup	NM_007294.4: c.3748G>T	—	—	NM_007194.4: c.1323_1324del	—	—	—	—	—	
Peru (n = 17)											
Positive	0	0	0	0	0	0	0	0	0	0	0
Variants	—	—	—	—	—	—	—	—	—	—	—
Overall[a] (n = 379)											
Positive	3 (0.8)	1 (0.3)	3 (0.8)	2 (0.5)	4 (1)	1 (0.3)	2 (0.5)	16 (4.2)	1 (0.3)	3 (0.8)	20 (5.27)

NOTE. Only mutations with at least one positive case shown.

Abbreviation: HRR, homologous recombination repair.

[a]There were missing samples in Argentina (two), Brazil (one), Colombia (two), and Costa Rica (three). Thus, the country and overall totals have been adjusted to reflect this and that is why they are slightly different from the CONSORT diagram totals.

TABLE 3. Associations According to Cramer's V Between Patient Characteristics and Germline Mutations

Characteristic Associated With HRR Germline Mutations	Cramer's V[a]	Subjects, No.
Age	0.25331	387
Country	0.20994	387
Family cancer history	0.05117	387
Medical history of cancer	0.01188	387
Location of metastases	0.12626	387
ECOG	0.22037	387

Abbreviations: ECOG, Eastern Cooperative Oncology Group; HRR, homologous recombination repair.
[a]Cramer's V interpretation: 0.1-0.39, weak positive association; 0.4-0.5, medium positive association; >0.5, strong positive association; −0.1 to −0.39, weak negative association; −0.4 to −0.5, medium negative association; <−0.5, strong negative association.

on the basis of their prognostic and predictive role.[10,14,15,18,19] In the past few years, several phase 3 trials demonstrated the role of targeted therapies in improving clinical outcomes of patients with mCRPC harboring HRR mutations, either as monotherapy or combined with novel hormonal therapies.[10,12,13,15] However, the data on the prevalence of germline and somatic HRR gene defects in LAC's patient population are largely unknown. For instance, in the pivotal phase 3 trials PROfound, TALAPRO-2, and PROpel, the representation of patients from LAC was 6%, 3%, and 18%, respectively.[10,12,13,15]

We conducted PROSPECT to understand and report the genomic alterations of patients with mCRPC in LAC, particularly the prevalence of HRR in this patient population. Other efforts are being conducted in other populations that are not well represented in the published literature, such as the Russian population (ClinicalTrials.gov identifier: NCT04712890) or Indian population (Clinical Trials Registry India registration code: CTRI/2021/12/038578).[20]

Our PROSPECT cohort included a good representation from all seven participating countries with more than 15 patients per country included; notably, more patients were enrolled from Mexico (23%). Patients had overall a low tumor burden on the basis of the low frequency of visceral metastases and multiple organ involvement. The low rate of previous local therapy suggests that many patients likely presented with newly diagnosed metastatic disease and were offered genetic testing within 1 year of developing mCRPC.

The prevalence of HRR germline mutations was lower than expected in all countries except in Brazil. We would expect HRR mutations to be present in at least 10% of patients, as shown in a large germline European study by Pritchard et al[8] and other studies.[21,22] Among the mutated genes, the proportion of *BRCA2*, *BRCA1*, and *ATM* mutations was concordant with previous reports from United States and Europe,[23] whereas the less frequent genes were underrepresented in this cohort. It is possible that the incidence of mutations in the rarer genes of the HRR family is different in patients from LAC and requires further investigation. In addition, there were frequent VUS in the HRR genes, which might yield pathogenic significance for patients with PC in LAC. These findings require further validation as these LAC populations have unique genetic compositions derived from Native American, European, and African ancestries that vary depending on the country and may underlie the low rate of HRR germline mutations detected in this study.[24]

Archival tissue was available in approximately two thirds of patients (64.8%, 251/387), concordant with other real-world data sets where tissue sampling is either unavailable or very old or poor quality. We noted that most tissue samples (79.3%, 199/251) were inadequate for genomic testing because of insufficient DNA, no tumor cells, or poor-quality DNA. The fact that most (84%, 212/251) of the archival tissue had been stored for ≤4 years does not seem to explain the very low rate of sample viability (<40%), which is approximately half of what is typically reported in clinical studies.[12,25-27] We observed in PROSPECT similar reasons for FFPE testing failures, including insufficient tumor tissue, exhausted FFPE blocks during the histologic diagnosis, insufficient tumor content to perform genomic tests, and poor-quality or insufficient DNA because of degradation during fixation or storage.[28] Importantly, the low rate of sample viability in PROSPECT was not associated with one specific pool of patients, country, or pathology laboratory.

Genetic testing rates are rising globally, yet still more than two thirds of patients are not being tested for HRR in the United States, despite several commercially available tests.[29] In LAC, the availability of genetic tests is limited and not available at all. Thus, PROSPECT was also the only opportunity to allow a significant number of patients to access genetic testing at no cost.

We acknowledge several limitations in the PROSPECT study. The available data on baseline disease and patient characteristics at the time of analysis were limited as a very simplified case report form was used in this light-touch study. Similarly, subsequent treatments and the clinical outcomes of the patient cohort were not the focus of this study, thus we were not able to assess the therapeutic implications of the genomic testing results and the use of genomic-based therapies such as poly ADP-ribose polymerase inhibitors. Importantly, at the time of study launch, circulating tumor DNA was not considered standard of care and was not assessed.

As we conducted this study, viable tumor archival tissue was identified as the main systematic barrier to tissue-based somatic testing in LAC. This finding led the sponsor of this study (AstraZeneca) to start preanalytical educational program for pathologists in collaboration with the College of American Pathologists that will be launched in July 2023 and until the first quarter of 2024 with a focus on sample acquisition, tissue optimization, and preservation. Similarly, some investigators from LAC are reporting better access to genetic testing, and liquid biopsy is now available in some countries such as Brazil and Colombia since early 2023.[30,31]

In conclusion, PROSPECT is one of the first studies to prospectively describe the prevalence of germline HRR mutations in patients with advanced PC from LAC, a largely underserved population. Germline HRR alterations were found at a lower rate than what has been observed in American and European cohorts. Viable tumor archival tissue was identified as the main systematic barrier to tissue-based somatic testing in LAC. On the basis of these findings, ongoing initiatives are being implemented to improve access to HRR testing for patients with mCRPC.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/po/author-center.
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